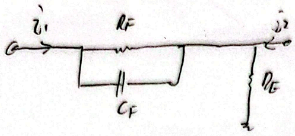
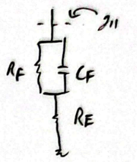
Derivation for Root Locus for Feedback-Zero Compensation



$$g_{12} = \frac{i_1}{i_2}\Big|_{v_1=0} = f$$

$$R_F \parallel \frac{1}{sC_F} = \frac{R_F \frac{1}{sC_F}}{R_F + \frac{1}{sC_F}} = \frac{R_F}{1+sR_FC_F}$$

$$g_{11} = \frac{i_1}{v_1}\Big|_{i_2=0}$$

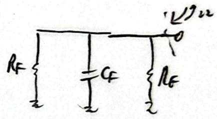


$$= -\frac{R_E}{R_E + R_F \parallel \frac{1}{sC_F}}$$

$$= -\frac{R_E}{R_E + \frac{R_F}{1+sR_FC_F}}$$

$$= -\frac{R_E + sR_ER_FC_F}{R_E + sR_ER_FC_F + R_F}$$

$$g_{22} = \frac{v_2}{i_2}\Big|_{v_1=0}$$



$$= -\frac{R_E}{R_F+R_E} \; \frac{1 + sR_FC_F}{1 + s\frac{R_ER_FC_F}{R_E+R_F}} \quad \#$$

$$g_{21} = \frac{v_2}{v_1}\Big|_{i_2=0}$$

$\Rightarrow$ neglected

$\hookrightarrow$ f has 1 zero & 1 pole

$$\omega_z = \frac{1}{R_FC_F} \qquad \omega_p = \frac{R_E+R_F}{R_E}\,\frac{1}{R_FC_F}$$

$$\frac{1}{\sigma+10} + \frac{1}{\sigma+20} = \frac{1}{\sigma+50}$$

$$(\sigma+50)(\sigma+20) + (\sigma+10)(\sigma+50) = (\sigma+10)(\sigma+20)$$

$$\sigma^2 + 70\sigma + 1000 + \cancel{\sigma^2} + 60\sigma + 500 = \cancel{\sigma^2} + 30\sigma + 200$$

$$\sigma^2 + 100\sigma + 1300 = 0$$

$$\sigma = -15.4\,,\ -84.6$$